SRI VASAVI ENGINEERING COLLEGE (Autonomous)

(Permanent Affiliation to JNTUK, Kakinada),

PEDATADEPALLI, TADEPALLIGUDEM-534 101

A.Y: 2024-25

III SEM CSE Handbook
(V23 Regulation)

Department of Computer Science and Engineering (Accredited by NBA)

Pedatadepalli, Tadepalligudem-534101, A.P

INDEX

INSTITUTE

VISION MISSION

INSTITUTE VISION AND MISSION

VISION

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

MISSION

- To produce engineering graduates of professional quality and global perspective through Learner Centric Education.
- To establish linkages with government, industry and research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.

DEPARTMENT

VISION

MISSION

DEPARTMENT

VISION AND MISSION

Vision:

- To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

Mission:

- To utilize innovative learning methods for academic improvement.
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character

PROGRAMME EDUCATIONAL OBJECTIVES, PROGRAMME OUTCOMES & PROGRAMME SPECIFIC OUTCOMES

Program Educational Objectives (PEOs): Graduates of this programme will

PEO 1: Adapt to evolving technology.

PEO 2: Provide optimal solutions to real time problems.

PEO 3: Demonstrate his/her abilities to support service activities with due consideration for Professional and Ethical Values.

Programme Specific Outcomes (PSO s): A graduate of the Computer Science and Engineering Program will be able to:

PSO 1: Use Mathematical Abstractions and Algorithmic Design along with Open Source Programming tools to solve complexities involved in Programming. [K3]

PSO 2: Use Professional engineering practices and strategies for development and maintenance of software. [K3]

Program Outcomes (POs): Computer Science Engineering Graduates will be able to:

1. **Engineering knowledge**: Apply the knowledge of Mathematics, Science, Engineering Fundamentals and Concepts of Computer Science Engineering to the solution of complex Engineering problems. **[K3]**

2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, Natural Sciences and Computer Science. **[K4]**

3. **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specific needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations. **[K5]**

4. **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. **[K5]**

5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex Engineering activities with an understanding of the limitations**.** **[K3]**

6. **The Engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional Engineering practice. **[K3]**

7. **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. **[K3]**

8. **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the Engineering practice. **[K3]**

9. **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. **[K6]**

10. **Communication**: Communicate effectively on complex Engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. **[K2]**

11. **Project management and finance**: Demonstrate knowledge and understanding of the Engineering and Management principles and apply these to one’s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. **[K6]**

12. **Life-long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. **[K1]**

ACADEMIC CALENDAR

✉ : principal@srivasaviengg.ac.in
svec.a8@gmail.com

☎: 08818- 284344, 355

SRI VASAVI ENGINEERING COLLEGE (AUTONOMOUS)
(Sponsored by Sri Vasavi Educational Society)
(Approved by AICTE, New Delhi & Permanently affiliated to JNTUK, Kakinada)
(Accredited by NAAC with 'A' Grade ,Recognized by UGC under section 2(f) & 12(B))
(NBA Accreditation to B.Tech., EEE,CSE, ME and ECE Branches for 3 Years)
Pedatadepalli, TADEPALLIGUDEM – 534 101. W.G.Dist. (A.P)

Principal's Office
Date: 03-07-2024

Academic Calendar
For II B.Tech (III and IV Semesters), Academic Year 2024-25

III Semester			
Description	**From**	**To**	**Weeks**
Commencement of Class Work	22.07.2024		
I Unit of Instructions	22.07.2024	14.09.2024	8 W
I Mid Examinations	16.09.2024	21.09.2024	1 W
II Unit of Instructions	23.09.2024	16.11.2024	8 W
II Mid Examinations	18.11.2024	23.11.2024	1 W
Preparation & Practicals	25.11.2024	30.11.2024	1 W
End Examinations	02.12.2024	14.12.2024	2 W
Commencement of Next Semester Class Work (IV Semester)	16.12.2024		
IV Semester			
I Unit of Instructions	16.12.2024	08.02.2025	8 W
I Mid Examinations	10.02.2025	15.02.2025	1 W
II Unit of Instructions	17.02.2025	12.04.2025	8 W
II Mid Examinations	14.04.2025	19.04.2025	1 W
Preparation & Practicals	21.04.2025	26.04.2025	1 W
End Examinations	28.04.2025	10.05.2025	2 W
Community Service Project	12.05.2025	05.07.2025	8 W
Commencement of Next Semester Class Work (V Semester)	07.07.2025		

PRINCIPAL

Copy to : ALL

Vision
To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

Mission
- To produce Engineering graduates of professional quality and global perspective through learner-centric education.
- To establish linkages with government, industry and Research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.

SRI VASAVI ENGINEERING COLLEGE (Autonomous)

Pedatadepalli, TADEPALLIGUDEM-534 101, W.G. Dist.

Department Of Computer Science &Engineering(Accredited by NBA)

III SEM CLASS CONSOLIDATED TIME TABLE w.e.f. 22.07.2024

Section – A Class Coordinator: Mr. P. Uma Sankar Room: B-301

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	DLCO	ADSAA	DMGT	MEFA	LUNCH BREAK	**PCS-I**		**Library**
Tue	ADSAA	**OOPs through Java**				MEFA	DLCO	ADSAA
Wed	**DTI**		OOPTJ			DMGT	MEFA	DLCO
Thu	OOPTJ	DMGT	ADSAA	DLCO		**ADSAA Lab**		
Fri	MEFA	DLCO	DMGT	ADSAA		OOPTJ	**DTI**	
Sat	ADSAA	OOPTJ	OOPTJ	DMGT		DLCO	MEFA	**Sports**

Section – B Class Coordinator: Mrs.A Naga Jyothi Room: B-302

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	ADSAA	**OOPs through Java**			LUNCH BREAK	DLCO	OOPTJ	DMGT
Tue	OOPTJ	DMGT	DLCO	ADSAA		MEFA	**DTI**	
Wed	DLCO	MEFA	DMGT	ADSAA		OOPJ	**DTI**	
Thu	MEFA	ADSAA	DLCO	OOPTJ		**PCS-I**		ADSAA
Fri	DMGT	MEFA	OOPTJ	DLCO		**ADSAA LAB**		
Sat	DLCO	DMGT	MEFA	**Library**		ADSAA	OOPTJ	**Sports**

Section – C Class Coordinator: Mr. P. Rama Mohan Rao Room: B-303

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	**DTI**		DLCO	ADSAA	LUNCH BREAK	DMGT	MEFA	OOPJ
Tue	DLCO	ADSAA	OOPTJ	MEFA		**ADSAA LAB**		
Wed	DMGT	MEFA	**PCS-I**			DLCO	ADSAA	MEFA
Thu	ADSAA	**OOPs through Java**				OOPTJ	DLCO	DMGT
Fri	MEFA	OOPTJ	OOPTJ	**Library**		DLCO	DMGT	ADSAA
Sat	**DTI**		OOPTJ	DLCO		DMGT	ADSAA	**Sports**

Section –D Class Coordinator: Mr. T Anil Kumar Reddy Room: B-304

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)		(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	DMGT	MEFA	**PCS-I**		LUNCH BREAK	OOPTJ	DLCO	ADSAA
Tue	**DTI**		DMGT	ADSAA		OOPTJ	DLCO	MEFA
Wed	ADSAA	DLCO	OOPTJ	MEFA		**OOPs through Java**		
Thu	MEFA	DLCO	ADSAA	DMGT		ADSAA	**DTI**	
Fri	DLCO	**ADSAA LAB**				DMGT	OOPTJ	**Library**
Sat	OOPTJ	ADSAA	DLCO	MEFA		MEFA	OOPTJ	**Sports**

Staff Details:

S. No.	Course Code	Course Name	Sections A	B	C	D
1.	V23MAT05	Discrete Mathematics & Graph Theory **(DMGT)**	Mr. G.Sriram Ganesh	Mr. G.Sriram Ganesh	Mr. G.Sriram Ganesh	Dr. V.S.Naresh
2.	V23MBT51	Managerial Economics and	Mr. R V Raja Sekhar	Dr. K Pulla Rao	Dr. K Rambabu	Mrs. V Sandhya
3.	V23CST03	Digital Logic & Computer Organization **(DLCO)**	Mr. P. Rama Mohan Rao	Mr. P. Rama Mohan Rao	Mr. P. Rama Mohan Rao	Ms. Y Divya Vani
4.	V23CST04	Advanced Data Structures & Algorithm	Mrs. B Sri Ramya	Mrs.A Naga Jyothi	Mr. K. Lakshmi Narayana	Mr. K. Lakshmi Narayana
5.	V23CST05	Object Oriented Programming Through Java **(OOPTJ)**	Mr. P. Uma Sankar	Mr. P. Uma Sankar	Mr. T Anil Kumar Reddy	Mr. T Anil Kumar Reddy
6.	V23CSL04	Advanced Data Structures and Algorithm Analysis Lab **(ADSAA Lab)**	Mrs. B Sri Ramya / Ms. T Pranusha	Mrs.A Naga Jyothi / Ms. Y Divya Vani	Mr. K. Lakshmi Narayana / Dr. G Tej Varma	Mr. K. Lakshmi Narayana / Ms.Vineela Mallina
7.	V23CSL05	Object Oriented Programming Through Java Lab **(OOPs through Java Lab)**	Mr. P. Uma Sankar / Mrs.M N V Surekha	Mr. P. Uma Sankar / Ms. G Nagavallika	Mr. T Anil Kumar Reddy / Ms. G Nagavallika	Mr. T Anil Kumar Reddy / Mrs. D S L Manikanteswari
8.	V23CSSE01	**Skill Enhancement Course:** Python Programming Lab **(PP Lab)**	Mrs. B Sri Ramya / Mrs. S Santhi Rupa	Dr. K Shirin Bhanu / Dr. G Tej Varma	Mr.N V Ratna Kishore Gade / Mr. L Atri Datha RaviTez	Mr. K. Lakshmi Narayana / Mr. T Anil Kumar Reddy
9.	V23MET09	Design Thinking & Innovation **(DTI)**	Mrs. S Santhi Rupa	Mrs. S Santhi Rupa	Mrs. S Santhi Rupa	Mrs. S Santhi Rupa
10.	V23ENT02	Professional Communication Skills - I **(PCS-I)**	Dr. K. Venkata Rao & Mr.Ch Mutyala Rao	Mr. M. Venkata Ramana & Mrs.Tanuja	Dr. B. Ananda Rao & Dr. K. Venkata	Mrs.P.V.Padmavathi & Mrs.Ch Manjeera

Lab Venues:

S.No.	Name of the Lab	Lab Venue
1	Advanced Data Structures and Algorithm Analysis Lab **(ADSAA Lab)**	James Gosling Lab (B Block Ground Floor)
2	Object Oriented Programming Through Java **(OOPs through Java Lab)**	

Head of the Department

Head of the Department
Dept. of Computer Science & Engineering
Sri Vasavi Engineering College
TADEPALLIGUDEM-534 101

COURSE STRUCTURE

SEMESTER-III (SECOND YEAR)

S.No.	Course Code	Name of the Course		L	T	P	C
1	V23MAT05	Discrete Mathematics & Graph Theory	BS&H	3	0	0	3
2	V23MBT51	Managerial Economics and Financial Analysis	Management Course- I	2	0	0	2
3	V23CST03	Digital Logic & Computer Organization	ESC	3	0	0	3
4	V23CST04	Advanced Data Structures & Algorithm Analysis	PCC	3	0	0	3
5	V23CST05	Object Oriented Programming Through Java	PCC	3	0	0	3
6	V23CSL04	Advanced Data Structures and Algorithm Analysis Lab	PCC	0	0	3	1.5
7	V23CSL05	Object Oriented Programming Through Java Lab	PCC	0	0	3	1.5
8	V23CSSE01	Python Programming Lab	Skill Enhancement Course	0	1	2	2
9	V23MET09	Design Thinking & Innovation	BS&H	1	0	2	2
10	V23ENT02	Professional Communication Skills - I	Audit Course	2	0	0	0
			Total	**17**	**1**	**10**	**21**

LESSON PLANS

Discrete Mathematics and Graph Theory

Academic Year: 2024-25 **Programme: B.Tech**
Year/ Semester: III **Section: A,B,C& D**
Name of the Course: Discrete Mathematics and Graph Theory
Course Code: V23MAT06

LESSON PLAN

Course Outcomes (Along with Knowledge Level): After completion of this course, Student will be able to:

S. No.	CO. No.	Course Outcome	BTL
1.	CO1	Develop skills required for solving mathematical problems using mathematical logic.	K3
2.	CO2	Demonstrate the set theory principles, relations and functions in real-time situations.	K3
3.	CO3	Apply the knowledge of combinatorics and recurrence relations in formulating and solving complex problems.	K3
4.	CO4	Apply the graph theory principles and techniques in computer science-related problems.	K3
5.	CO5	Find shortest paths and minimal spanning trees using prim's and Kruskal's algorithms, BFS and DFS algorithms.	K3

Text Books:

1. Discrete Mathematical Structures with Applications to Computer Science, J.P. Tremblay and P. Manohar, Tata McGraw Hill.
2. Elements of Discrete Mathematics-A Computer Oriented Approach, C.L.LiuandD. P. Mohapatra, 3rdEdition, TataMcGraw Hill.
3. The ory and Problems of Discrete Mathematics, Schaum's Outline Series, Seymour Lipschutz and Marc Lars Lipson, 3rd Edition, McGraw Hill.

Reference Books:

1. Discrete Mathematics for Computer Scientists and Mathematicians, J.L.Mott, A.Kandel and T. P. Baker, 2nd Edition, Prentice Hall of India.
2. Discrete Mathematical Structures, Bern and Kolman, Robert C. Bus by and Sharon Cutler Ross, PHI.
3. Discrete Mathematics, S.K. Chakraborthy and B.K.Sarkar, Oxford,2011.
4. Discrete Mathematics and its Applications with Combinatorics and Graph Theory, K.H.Rosen,7th Edition,Tata Mc GrawHill.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		60	60	60	60	60
Targeted level of Attainment	**Level 3**	60	60	60	60	60
	Level 2	50	50	50	50	50
	Level 1	40	40	40	40	40

Lecture Plan:

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO1	Dissemination of Vision, Mission, PEOs, POs,PSOs		1	Lecture	IFP
2		**Mathematical Logic:** Define Statements and their Notations, Connectives	K1	1	Lecture	IFP
3		Describe Well Formed Formulas, Truth Tables, Tautologies	K2	1	Lecture with Discussion	IFP
4		Explain equivalence of Formulas	K2	2	Lecture	IFP
5		State duality Law, Tautological implications	K1	1	Lecture with Discussion	IFP
6		Explain normal forms	K2	2	Lecture	IFP
7		Illustrate theory of inference for statement calculus	K3	2	Lecture	IFP
8		Practice Consistency of Premises , Indirect method of proof	K3	2	Lecture	IFP
9		**Predicate Calculus:** Identify Predicates, Predicative Logic, Statement Functions	K2	1	Lecture with Discussion	IFP
10		Recognize Variables and Quantifiers, Free and Bound Variables	K2	2	Lecture	IFP
11		Illustrate Inference Theory for Predicate Calculus.	K3	2	Lecture	IFP
		Total		**17**		

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO 2	**Set Theory and Relations:** Define basic concepts	K1	1	Lecture	IFP
2		Illustrate operations on binary sets	K2	1	Lecture	IFP
3		Use principle of inclusion and exclusion	K3	1	Lecture	IFP
4		Describe Relation and properties of binary relations on a set and Transitive Closure	K2	2	Lecture	IFP
5		Sketch out relation matrix and digraph	K3	1	Lecture with Discussion	IFP

6		Practice equivalence, compatibility, and partial ordering relations	K3	2	Lecture with Discussion	IFP
7		Construct hasse diagrams, lattice and state its properties.	K3	2	Lecture	IFP
8		**Functions:** Illustrate Bijective, Composition, Inverse, Permutation, and Recursive Functions	K3	3	Lecture	IFP
		Total		**13**		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO3	**Combinatorics:** Explain Basis of Counting	K2	1	Lecture	IFP
2		Solve Permutations, Permutations with Repetitions	K3	1	Lecture with discussion	IFP
3		Solve Circular and Restricted Permutations	K3	1	Lecture with discussion	IFP
4		Solve Combinations, Restricted Combinations	K3	1	Lecture with discussion	IFP
5		Discuss Binomial and Multinomial Coefficients and Theorems	K2	2	Lecture with discussion	IFP
6		**Recurrence Relations** Explain Generating of functions	K2	2	Lecture	IFP
7		Calculate Coefficient of generating functions	K3	2	Lecture	IFP
8		Explain Recurrence relations	K2	1	Lecture with discussion	IFP
9		Solve homogeneous Recurrence relations by method of substitution	K3	1	Lecture	IFP
10		Solve homogeneous Recurrence relations by Generating functions	K3	2	Lecture with discussion	IFP
11		Solve Recurrence relations by method of characteristic roots	K3	1	Lecture with discussion	IFP
12		Solve inhomogeneous recurrence relations	K3	2	Lecture	IFP
		Total		**17**		

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO 4	Describe basic concepts of graphs	K1	1	Lecture with Discussion	IFP
2		Explain Graph Theory and its Applications	K2	1	Lecture with Discussion	IFP
3		Illustrate matrix representation of graphs, Adjacency matrices, Incidence matrices	K2	1	Lecture	IFP
4		Find subgraph, isomorphic graphs, paths and circuits	K3	2	Lecture	IFP
5		Demonstrate Eulerian and Hamiltonian Graphs,	K3	2	Lecture with Discussion	IFP
6		Total		**7**		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO 5	Explain Multi Graphs, Bipartite and Planar Graphs.	K1	1	Lecture	IFP
2		Use Euler's Formula for Planar Graphs	K3	1	Lecture	IFP
3		Explain Graph Colouring and Chromatic Number	K2	1	Lecture	IFP
4		Explain tree and spanning trees	K2	2	Lecture	IFP
5		Sketch Minimal spanning trees using Kruskal's algorithms	K3	2	Lecture	IFP
6		Sketch Minimal spanning trees using Prim's algorithms	K3	2	Lecture	IFP
7		Construct BFS	K3	2	Lecture	IFP
8		Construct DFS	K3	2	Lecture	IFP
		Total		**13**		

Total No. of Classes: 67

Managerial Economics and Financial Analysis

Academic Year: 2024-25 **Programme: B.Tech**
Year/ Semester: III **Section: A,B,C& D**
Name of the Course: Managerial Economics and Financial Analysis
Course Code: V23MBT51

LESSON PLAN

Course Outcomes (Along with Knowledge Level): After completion of this course, Student will be able to:

S. No	CO. No	Course Outcomes	BTL
1	CO1	Understanding the basic concepts of managerial economics, demand, elasticity of demand and methods of demand forecasting	**K2**
2	CO2	Interpret production concept, least cost combinations and various costs concepts in decision making.	**K3**
3	CO3	Differentiate various Markets and Pricing methods along with Business Cycles.	**K2**
4	CO4	Prepare financial statements and its analysis.	**K3**
5	CO5	Assess various investment project proposals with the help of Capital Budgeting techniques for decision making	**K3**

Text Books:

1. Dr. N. Appa Rao, Dr. P. Vijay Kumar: 'Managerial Economics and Financial Analysis', Cengage Publications, New Delhi – 20112.
2. Dr. A. R. Aryasri – Managerial Economics and Financial Analysis, TMH 2011.

Reference Books:

1 Dr. B. Kuberudu and Dr. T. V. Ramana: Managerial Economics &Financial Analysis,Himalaya Publishing House, 2014

2. S. A. Siddiqui; A. S. Siddiqui: Managerial Economics and Financial Analysis, New AgeInternational Publishers, 2012.

Targeted Proficiency and attainment Levels (for each Course Outcome):

COs		**CO1**	**CO2**	**CO3**	**CO4**	**CO5**
Targeted Proficiency Level		60	60	60	60	60
Targeted level of Attainment	**Level 3**	60	60	60	60	60
	Level 2	50	50	50	50	50
	Level 1	40	40	40	40	40

Lecture Plan:

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO1	Define managerial economics	K1	1	Lecture Discuss	IFP
2		Describe ME with other disciplines	K1	1	Lecture	IFP
3		Explain Nature and scope of managerial economics	K2	1	Lecture	IFP
4		Define Demand	K1	1	Lecture Discuss	IFP
5		Describe law of demand	K2	1	Lecture	IFP
6		Explain Elasticity of demand	K2	2	Lecture	IFP
7		Find the of elasticity of demand	K2	2	Lecture	IFP
8		Explain Demand forecasting, methods.	K2	2	Lecture Discuss	IFP
		Total		11		
S.No	**Course Outcome**	**Intended Learning Outcomes (ILO)**	**Knowledge Level of ILO**	**No. of Hours Required**	**Pedagogy**	**Teaching aids**
1	CO2	State Production function	K1	1	Lecture	IFP
2		State Isocost	K1	1	Lecture	IFP
3		State Iso quants	K1	1	Lecture	IFP
4		Explain Cob-Douglas production function	K2	1	Lecture Discuss	IFP
5		Describe economies of scale	K2	1	Lecture	IFP
		Enumerate various cost concepts	K1	1	Lecture	IFP
		Solve break even analysis problems	K3	4	Lecture	IFP
		Total		**10**		
1	CO3	Describe Different types of market structures	K1	3	Lecture	IFP
2		Explain Price-output determination under different market structures	K2	4	Lecture	IFP
3		Explain Pricing objectives, Cost and demand based Pricing methods	K2	2	Lecture	IFP

4		Describe competition, strategy based pricing methods.	K2	2	Lecture	IFP
5		State the meaning and features of business cycles	K1	1	Lecture	IFP
6		Describe the Phases of business Cycles.	K2	2	Lecture	IFP
		Total		**14**		
1	**CO4**	Describe double entry system	K2	3	Lecture	IFP
2		Preparation of financial statements	K3	4	Lecture	IFP
3		Interpretation of financial statements by using, Ratios.	K3	6	Lecture	IFP
		Total		**13**		
1	**CO5**	Define Capital	K1	1	Lecture	IFP
2		Explain significance of of capital budgeting	K1	1	Lecture	IFP
3		Explain capital budgeting, Process	K2	4	Lecture	IFP
5		Apply capital budgeting techniques	K3	4	Lecture	IFP
		Total		**10**		

Total No. of Classes: 58

DIGITAL LOGIC & COMPUTER ORGANIZATION

Academic Year: 2024-25 **Programme: B.Tech**
Year/ Semester: III **Section: A,B,C& D**
Name of the Course: DIGITAL LOGIC & COMPUTER ORGANIZATION
Course Code: V23CST03

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level): After completion of this course, the students will be able to:

S. No.	CO No.	Course Outcome	BTL
1.	CO1	Explain different Data Representation and various Combinational Digital Logic Circuits.	K2
2.	CO2	Explain various Sequential Digital Logic Circuits and Basic Structure of Computers.	K2
3.	CO3	Describe the Computer Arithmetic and basic concepts of Processor Organization	K2
4.	CO4	Illustrate different types of Memory.	K2
5.	CO5	Demonstrate the ways of accessing various Interfacing devices with processor.	K3

Text Books:

1. Computer Organization, Carl Hamacher, ZvonkoVranesic, SafwatZaky, 6th edition,
2. Digital Design, 6th Edition, M. Morris Mano, Pearson Education.
3. Computer Organization and Architecture, William Stallings, 11thEdition, Pearson.

Reference Books:

1. Computer Systems Architecture, M.Moris Mano, 3rdEdition, Pearson.
2 Computer Organization and Design, David A. Paterson, John L.Hennessy, Elsevier
3. Fundamentals of Logic Design, Roth, 5thEdition, Thomson.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		**CO1**	**CO2**	**CO3**	**CO4**	**CO5**
Targeted Proficiency Level		60	60	60	60	60
Targeted level of Attainment	**Level 3**	60	60	60	60	60
	Level 2	50	50	50	50	50
	Level 1	40	40	40	40	40

Lecture Plan:

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Dissemination of Vision, Mission of the Dept. and PEOs, Pos, & PSOs of the Programme		1	Lecture	IFP
		Data Representation: Digital Logic Circuits-I:				
2		Explain Binary Numbers.	K2	3	Lecture with Discussion	IFP
3		Explain Fixed Point Representation.	K2	1	Lecture with Discussion	IFP
4		Explain Floating Point Representation.	K2	1	Lecture with Discussion	IFP
5		Describe Signed binary umbers.	K1	1	Lecture with Discussion	IFP
6		Explain Basic Logic Functions, Logic gates	K2	2	Lecture with Discussion	IFP
7		Explain universal logic gates,	K2	1	Lecture with Discussion	IFP
8		Describe Minimization of Logic expressions. K-Map Simplification.	K1	2	Lecture with Discussion	IFP
9		Illustrate the Combinational Circuits, Decoders, Multiplexers	K2	2	Lecture with Discussion	IFP
		Total		**14**		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
		Digital Logic Circuits-II: Basic Structure of Computers:				
10	CO 2	Explain Sequential Circuits, Flip-Flops,	K2	2	Lecture with Discussion	IFP
11		Explain Binary counters, Registers	K2	2	Lecture with Discussion	IFP
12		Explain Shift Registers, Ripple counters	K2	2	Lecture	IFP
13		Explain Computer Types, Functional units, Basic operational concepts	K2	2	Lecture with Discussion	IFP
14		Explain Bus structures, Software, Performance	K2	2	Lecture with Discussion	IFP
15		Explain multiprocessors and multi computers.	K2	1	Lecture with Discussion	IFP
16		Explain Computer Generations, Von- Neumann Architecture	K2	2	Lecture with Discussion	IFP
		Total		13		

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
		Computer Arithmetic : Processor Organization:				
17	CO 3	Describe The Addition and Subtraction of Signed Number	K1	1	Lecture with Discussion	IFP
18		Explain Design of Fast Adders, Multiplication of Positive Numbers	K2	2	Lecture with Discussion	IFP
19		Explain Signed-operand Multiplication, Fast Multiplication, Integer Division,	K2	2	Lecture with Discussion	IFP
20		Describe Floating-Point Numbers and Operations	K1	1	Lecture with Discussion	IFP
21		Describe the Fundamental Concepts in execution of instruction.	K1	1	Lecture	IFP
22		Describe the Execution of instruction involves register transfer.	K1	1	Lecture with Discussion	IFP
23		Explain the multiple bus organization.	K2	1	Lecture with Discussion	IFP
24		Illustrate the hardwired control unit.	K2	1	Lecture with Discussion	IFP
25		Illustrate the micro programmed control unit.	K2	2	Lecture with Discussion	IFP
		Total		**12**		

S .No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
		The Memory Organization:				
26	**CO 4**	Describe Basic Concepts of Memory.	K1	1	Lecture	IFP
27		Discuss Semi Conductor RAM Memories.	K2	1	Lecture with Discussion	IFP
28		Discuss Read Only Memories.	K2	1	Lecture with Discussion.	IFP
29		Describe Speed, Size and Cost of memories.	K1	1	Lecture with Discussion	IFP
30		Illustrate the cache memory organization of computer	K2	2	Lecture with Discussion	IFP
31		Explain-Performance Considerations	K2	1	Lecture with Discussion	IFP
32		Illustrate Virtual Memories	K2	2	Lecture with Discussion	IFP
33		Explain Memory-Management Requirements.	K2	1	Lecture with Discussion	IFP
34		Explain Secondary Storage	K2	2	Lecture with Discussion	IFP
		Total		**12**		

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
		Input/Output Organization:				
35	**CO 5**	Describe the Accessing of I/O Devices.	K1	1	Lecture	BB/ICT
36		Demonstrate interrupt of I/O Devices.	K2	2	Lecture with Discussion	BB/ICT
37		Illustrate Processor Examples	K2	1	Lecture	IFP
38		Demonstrate Direct Memory Access	K2	2	Lecture with Discussion	IFP
39		Explain the Buses and its types.	K2	2	Lecture with Discussion	IFP
40		Illustrate Interface Circuits	K2	1	Lecture with Discussion	IFP
41		Illustrate Standard I/O Interfaces	K2	2	Lecture with Discussion	IFP
		Total		**11**		

Total No. of Classes: 63

Advanced Data Structures and Algorithm Analysis

Academic Year: 2024-25 **Programme: B.Tech**
Year/ Semester: III **Section: A,B,C& D**
Name of the Course: Advanced Data Structures and Algorithm Analysis
Course Code: V23CST04

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level): After completion of this course, the students will be able to:

S. No	CO. No	Course Outcomes	BTL
1	CO1	Demonstrate asymptotic notations and nonlinear data structures like AVL Trees and B-Trees	**K3**
2	CO2	Demonstrate graphs and Divide and conquer technique	**K3**
3	CO3	Use Greedy and Dynamic programming techniques to determine various problems	**K3**
4	CO4	Develop algorithms using Backtracking and Branch & Bound techniques	**K3**
5	CO5	Solve different graph problems using NP Hard and NP Complete Problems.	**K3**

Text Books:

1. Fundamentals of Data Structures in C++, Horowitz, Ellis; Sahni, Sartaj; Mehta, Dinesh, 2ndEdition Universities Press.
2. Computer Algorithms in C++, Ellis Horowitz, SartajSahni, SanguthevarRajasekaran, 2nd Edition University Press

Reference Books:

1. Data Structures and program design in C, Robert Kruse, Pearson Education Asia
2. An introduction to Data Structures with applications, Trembley& Sorenson, McGraw Hill
3. The Art of Computer Programming, Vol.1: Fundamental Algorithms, Donald E Knuth, Addison-Wesley, 1997.
4. Data Structures using C & C++: Langsam, Augenstein&Tanenbaum, Pearson, 1995.
5. Algorithms + Data Structures & Programs:, N.Wirth, PHI.
6. Fundamentals of Data Structures in C++: Horowitz Sahni& Mehta, Galgottia Pub.
7. Data structures in Java:, Thomas Standish, Pearson Education Asia

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
V23CST04.1	60	65
V23CST04.2	60	65
V23CST04.3	60	60
V23CST04.4	60	60
V23CST04.5	60	60

LESSON PLAN

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Describe Asymptotic Notations with examples	K2	2	Lecture+ Discussion	IFP
2		Explain insertion and deletion in AVL Trees	K2	3	Lecture+ Discussion	
3		Construct AVL Trees	K3	1	Lecture+	
4		Discuss Applications of AVL Trees	K2	1	Discussion	
5		Explain insertion and deletion in B Trees	K2	2	Lecture	
6		Construct B Trees of different orders	K3	1	Lecture+ Discussion	
		Total		**10**		

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO2	Describe Min and Max Heaps	K2	1	Lecture+ Discussion	IFP
2		Illustrate operations and applications of Min and Max Heaps	K3	2	Lecture+ Discussion	
3		Discuss Graph terminology, representations, Search & Traversals	K2	4	Lecture+ Discussion	
4		Explain Connected and Bi connected Components	K2	3	Lecture	
5		Use Divide and Conquer in General Method, Quick Sort and Merge Sort	K3	4	Lecture+ Discussion	
6		Illustrate Strassen's Matrix Multiplication and Convex Hull	K3	3	Lecture	
		Total		**17**		

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO3	Discuss Greedy and Dynamic Programming techniques	K2	1	Lecture	IFP
2		Illustrate Job Sequencing with deadlines	K3	2	Lecture+ Discussion	
3		Explain Knapsack Problem	K2	2	Lecture+ Discussion	
4		Construct Minimum Cost Spanning Trees using Prims and Kruskals algorithms	K2	3	Lecture	
5		Find single source shortest path using Dijkstra's algorithm	K2	2	Lecture	
		Total		**10**		

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO4	Solve Complex problems using General Method	K3	2	Lecture + Discussion	IFP
2		Illustrate Single Source Shortest Paths – General Weights (Bellman-Ford Algorithm)	K3	2	Lecture + Discussion	
3		Construct Optimal Binary Search Trees	K3	1	Lecture + Discussion	
4		Explain 0/1 Knapsack problem	K2	2	Lecture + Discussion	
5		Describe String Editing	K2	1	Lecture + Discussion	
6		Illustrate Travelling Salesperson Problem (TSP)	K3	2	Lecture + Discussion	
7		Illustrate General Method in backtracking,	K3	1	Lecture + Discussion	
8		Illustrate 8-Queens Problem with example	K3	2	Lecture + Discussion	
9		Illustrate Sum of Subsets Problem,	K3	1	Lecture + Discussion	
10		Explain Graph Coloring with example	K2	1	Lecture + Discussion	
		Total		**15**		

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Illustrate NP Hard and NP Complete Problems	K2	3	Lecture + Discussion	IFP
2		Explain Cook's Theorem	K3	2	Lecture + Discussion	
3		Illustrate NP Hard Graph Problems: CDP, CNDP, TSP	K3	4	Lecture+ Discussion	
4		Illustrate NP Hard Scheduling Problems	K3	2	Lecture+ Discussion	
5		Illustrate Job Shop scheduling with example	K3	2	Lecture+ Discussion	
		Total		**13**		

Total No. of Classes: 65

Object Oriented Programming Through Java

Academic Year: 2024-25 **Programme: B.Tech**
Year/ Semester: III **Section: A,B,C& D**
Name of the Course: Object Oriented Programming Through Java
Course Code: V23CST05

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level): After completion of this course, the students will be able to:

S. No.	CO No.	Course Outcome	BTL
1.	CO1	Demonstrate the object-oriented programming principles with Java programming environment.	K3
2.	CO2	Demonstrate the concepts like classes, objects, argument passing mechanism, overloading, and overriding.	K3
3.	CO3	Illustrate the concepts of arrays, inheritance, and interfaces.	K3
4.	CO4	Demonstrate packages, java libraries, exception handling, java I/O, and File concepts.	K3
5.	CO5	Illustrate the concepts of string handling, multithreading, and Java FX GUI.	K3

Textbooks:

1. JAVA one step ahead, Anitha Seth, B.L.Juneja, Oxford.
2. Joy with JAVA, Fundamentals of Object Oriented Programming, DebasisSamanta, Monalisa Sarma, Cambridge, 2023..
3. JAVA 9 for Programmers, Paul Deitel, Harvey Deitel, 4th Edition, Pearson.

Reference Books:

1. The complete Reference Java, 11thedition, Herbert Schildt,TMH
2. Introduction to Java programming, 7th Edition, Y Daniel Liang, Pearson

Targeted Proficiency and attainment Levels (for each Course Outcome):

COs	CO1	CO2	CO3	CO4
Targeted Proficiency Level	60	60	60	60
Targeted level of Attainment Level 3	60	60	60	60
Targeted level of Attainment Level 2	55	55	55	55
Targeted level of Attainment Level 1	50	50	50	50

Lecture Plan:

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching Aids
1	CO1	Syllabus, Cos and POs Introduction to Object-Oriented Programming	K2	1	Lecture with Discussion	IFP
2		Explain Java's Features and its History	K2	1	Lecture	IFP
3		Discuss Java Program Structure and Syntax	K2	1	Lecture with Examples	IFP
4		Describe Data Types, Variables, and Operators	K2	2	Lecture with Examples	IFP
5		Explain Control Statements and Looping Constructs	K2	2	Lecture with Discussion	IFP
6		Discuss Command Line Arguments and User Input Handling	K2	2	Lecture	IFP
7		Illustrate the Concept of Static Variables and Methods	K2	1	Lecture	IFP
8		Explain Java's Memory Management and Garbage Collection	K2	2	Lecture with Discussion	IFP
9		Demonstrate Basic Java Programs	K3	3	Practical Session	IFP
10		Discuss Programming Style and Best Practices	K2	1	Lecture	IFP
11		Explain Type Casting and Type Promotion	K2	2	Lecture with Examples	IFP
		Total		18		

S. No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching Aids
1	CO2	Introduction to Classes and Objects	K2	1	Lecture	IFP
2		Describe Class Declaration and Object Instantiation	K2	2	Lecture with Examples	IFP
3		Explain Constructors and Constructor Overloading	K2	2	Lecture with Discussion	IFP
4		Discuss Method Overloading and Overriding	K2	2	Lecture with Examples	IFP
5		Explain the use of the this Keyword	K2	1	Lecture	IFP
6		Illustrate Access Modifiers and Access Control	K2	1	Lecture	IFP
7		Discuss Passing Arguments by Value and Reference	K2	2	Lecture with Examples	IFP
8		Explain Nested and Inner Classes	K2	1	Lecture	IFP
9		Demonstrate Object Cloning and Copying Objects	K3	2	Practical Session	IFP

10		Discuss Final Classes and Methods	K2	1	Lecture	IFP
11		Illustrate Real-world Applications of Classes and Objects	K3	2	Practical Session	IFP
		Total		**17**		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching Aids
1	CO3	Introduction to Arrays	K2	1	Lecture	IFP
2		Discuss Array Declaration and Initialization	K2	2	Lecture with Examples	IFP
3		Explain Two-dimensional and Multidimensional Arrays	K2	2	Lecture with Examples	IFP
4		Illustrate Array Operations and Methods	K2	2	Lecture	IFP
5		Explain the Concept of Inheritance and its Types	K2	2	Lecture with Examples	IFP
6		Discuss Method Overriding and Dynamic Method Dispatch	K2	2	Lecture with Discussion	IFP
7		Illustrate the Use of the super Keyword and Constructor Chaining	K2	2	Lecture with Examples	IFP
8		Explain Interfaces and Abstract Classes	K2	2	Lecture with Examples	IFP
9		Discuss Implementing Multiple Interfaces	K2	1	Lecture	IFP
10		Demonstrate Real-world Applications of Inheritance and Interfaces	K3	2	Practical Session	IFP
		Total		**18**		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching Aids
1	CO4	Introduction to Packages and Access Control	K2	1	Lecture	IFP
2		Discuss Importing and Creating Packages	K2	2	Lecture with Examples	IFP
3		Explain Java's Built-in Packages (java.lang, java.util, etc.)	K2	2	Lecture	IFP
4		Discuss Exception Handling Mechanisms	K2	2	Lecture with Examples	IFP
5		Explain the Use of try, catch, and finally Blocks	K2	2	Lecture	IFP
6		Illustrate Custom Exceptions and Exception Hierarchies	K2	2	Lecture with Discussion	IFP

7		Explain Java I/O and File Handling Concepts	K2	2	Lecture with Examples	IFP
8		Discuss Byte Streams and Character Streams	K2	2	Lecture with Examples	IFP
9		Demonstrate File I/O Operations	K3	2	Practical Session	IFP
		Total		**17**		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching Aids
1	CO5	Introduction to Strings and String Handling	K2	1	Lecture	IFP
2		Discuss String Methods and Operations	K2	2	Lecture with Examples	IFP
3		Explain StringBuffer and StringBuilder Classes	K2	2	Lecture with Discussion	IFP
4		Illustrate Regular Expressions and Pattern Matching	K2	2	Lecture	IFP
5		Explain Multithreading and Concurrency Concepts	K2	2	Lecture with Examples	IFP
6		Discuss Creating Threads and Thread Lifecycle	K2	2	Lecture	IFP
7		Illustrate Synchronization and Inter-thread Communication	K2	2	Lecture with Discussion	IFP
8		Explain Java FX GUI Development	K2	2	Lecture with Examples	IFP
9		Discuss Event Handling in Java FX	K2	1	Lecture	IFP
10		Demonstrate Building a GUI Application with Java FX	K3	1	Practical Session	IFP
		Total		**17**		

Total No. of Classes: 87

Advanced Data Structures and Algorithm Analysis Lab

Academic Year: 2024-25 **Programme: B.Tech**
Year/ Semester: III **Section: A,B,C& D**
Name of the Course: Advanced Data Structures and Algorithm Analysis Lab
Course Code: V23CSL04

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level): After completion of this course, the students will be able to:

S.No.	CO No.	Course Outcome	BTL
1.	CO1	Demonstrate programs on AVL Trees and Heap trees.	K3
2.	CO2	Develop programs on Sorting algorithms and Graph traversal algorithms.	K3
3.	CO3	Develop programs using Greedy and Dynamic programming technique.	K3
4.	CO4	Develop programs using Backtracking and branch & Bound technique.	K3

Reference Books:

1. Fundamentals of Data Structures in C++, Horowitz Ellis, SahniSartaj, Mehta, Dinesh, 2^{nd}Edition, Universities Press
2. Computer Algorithms/C++ Ellis Horowitz, SartajSahni, SanguthevarRajasekaran, 2^{nd}Edition, University Press
3. Data Structures and program design in C, Robert Kruse, Pearson Education Asia
4. An introduction to Data Structures with applications, Trembley& Sorenson, McGraw Hill

Targeted Proficiency and attainment Levels (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
V23CSL04.1	60	65
V23CSL04.2	60	65
V23CSL04.3	60	60
V23CSL04.4	60	60

Lecture Plan:

S.No.	Course Outcome	Program Name	Knowledge Level	No. of Hours	Pedagogy	Teaching Aids
1	**CO1**	Construct an AVL tree for a given set of elements which are stored in a file. And implement insert and delete operation on the constructed tree. Write contents of tree into a new file using in-order.	K3	6	Lecture& Experiment	IFP
2		Construct B-Tree an order of 5 with a set of 100 random elements stored in array. Implement searching, insertion, and deletion operations.				
3	**CO2**	Construct Min and Max Heap using arrays, delete any element, and display the content of the Heap.	K3	12	Lecture& Experiment	IFP
4		Demonstrate BFT and DFT for given graph when graph is represented by a) Adjacency Matrix b) Adjacency Lists.				
5		Develop a program for finding the bi connected components in a given graph.				
6		Implement Quick sort and Merge sort and observe the execution time for various input sizes (Average, Worst, and Best cases).				
7		Examine the performance of Single Source Shortest Paths using Greedy method when the graph is represented by adjacency matrix and adjacency lists.				
8	**CO3**	Demonstrate Job Sequencing with deadlines using Greedy strategy.	K3	6	Lecture& Experiment	IFP
9		Demonstrate a program to solve 0/1 Knapsack problem using Dynamic Programming.				
10	**CO4**	Demonstrate N-Queens Problem using Backtracking.	K3	9	Lecture& Experiment	IFP
11		Use Backtracking strategy to solve 0/1 Knapsack problem.				
12		Demonstrate Travelling Sales Person problem using Branch and Bound approach.				

Total No. of Hours: 33

Object Oriented Programming through Java Lab

Academic Year: 2024-25 **Programme: B.Tech**
Year/ Semester: III **Section: A,B,C& D**
Name of the Course: Object Oriented Programming through Java Lab
Course Code: V23CSL05

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level): After completion of this course, the students will be able to:

S. No.	CO No.	Course Outcome	BTL
1.	CO1	Construct programs to handle classes and objects.	[K3]
2.	CO2	Develop programs that incorporate inheritance and interfaces.	[K3]
3.	CO3	Construct programs on exception handling and File I/O.	[K3]
4.	CO4	Develop programs using multithreading and Java FX.	[K3]

Text Books:

1. JAVA one step ahead, Anitha Seth, B.L. Juneja, Oxford.
2. Joy with JAVA, Fundamentals of Object Oriented Programming, Debasis Samanta, Monalisa Sarma, Cambridge, 2023.
3. JAVA 9 for Programmers, Paul Deitel, Harvey Deitel, 4th Edition, Pearson.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

COs	CO1	CO2	CO3	CO4
Targeted Proficiency Level	80	80	80	80
Targeted level of Attainment Level 3	75	75	75	75
Targeted level of Attainment Level 2	70	70	70	70
Targeted level of Attainment Level 1	65	65	65	65

Lecture Plan:

Exp. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Develop a JAVA program to display default values of all primitive data types of JAVA.	K3	15	Demonstration & Experiment	ICT
2		Develop a JAVA program that displays the roots of a quadratic equation $ax^2+bx+c=0$. Find the dis criminant D and basing on value of D, describe the nature of the root				ICT
3		Develop a JAVA program to search for an element in a given list of elements using binary search.				ICT
4		Develop a JAVA program to sort the list of elements using bubble sor				ICT
5		Develop a JAVA program using String Buffer to delete, remove character.				ICT
6		Develop a JAVA program to implement class Mechanism. Build a class, methods and invoke them inside main method.				ICT
7		Develop a JAVA program to implement method overloading.				ICT
8		Develop a JAVA program to implement constructor.				ICT
9		Develop a JAVA program to implement constructor overloading.				ICT
10	CO2	Develop a JAVA program to implement Single Inheritance.	K3	9	Demonstration & Experiment	ICT
11		Develop a JAVA program to implement multi level Inheritance.				ICT
12		c) Develop a JAVA program for abstract class to find areas of different shapes.				ICT
13		a) Develop a JAVA program on ―super‖ keyword.				ICT
14		b) Develop a JAVA program to implement Interface. What kind of Inheritance can be achieved?				ICT
15		c) Develop a JAVA program that implements Runtime polymorphism.				ICT

16	CO3	Exercise – 6: a) Develop a JAVA program that describes exception handling mechanism	K3	9	Demonstration & Experiment	ICT
17		Develop a JAVA program on Multiple catch clauses.				ICT
18		Develop a JAVA program to generate Built-in Exceptions.				ICT
19		Develop a JAVA program to generate User Defined Exception.				ICT
20	CO4	Develop a JAVA program that creates threads by extending Thread class. First thread display Good Morning —every 1 sec, the second thread displays —Hello —every 2 seconds and the third display —Welcome‖ every 3 seconds,(Repeat the same by implementing Runnable)	K3	15	Demonstration & Experiment	ICT
21		b) Develop a program on isAlive() and join ().				ICT
22		c) Demonstrate Daemon Threads.				ICT
23		Demonstrate Producer Consumer Problem.				ICT
24		a) Develop a JAVA program that import and use the user defined packages.				ICT
25		b) Demonstrate a GUI that display text in label and image in an ImageView (use JavaFX) Without writing any code.				ICT
26		Develop a Tip Calculator app using several JavaFX components and learn how to respond to user interactions with the GUI				ICT

Total No. of Classes:48

Design Thinking and Innovation

Academic Year: 2024-25 **Programme: B.Tech**
Year/ Semester: III **Section: A,B,C& D**
Name of the Course: Design Thinking and Innovation **CourseCode: V23MET09**

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level): After completion of this course, the students will be able to:

S. No	CO. No	Course Outcomes	BTL
1	CO1	Bring awareness on innovative design and new product development.	**K1**
2	CO2	Explain the basics of design thinking.	**K2**
3	CO3	Familiarize the role of reverse engineering in product development.	**K3**
4	CO4	Train how to identify the needs of society and convert into demand.	**K4**
5	CO5	Introduce product planning and product development process.	**K5**

Text Books:

1. Tim Brown, Change by design, 1/e, Harper Bollins, 2009.
2. Idris Mootee, Design Thinking for Strategic Innovation, 1/e, Adams Media, 2014.

Reference Books:

1. David Lee, Design Thinking in the Classroom, Ulysses press, 2018.
2. Shrrutin N Shetty, Design the Future, 1/e, Norton Press, 2018.
3. William lidwell, Kritinaholden, &Jill butter, Universal principles of design, 2/e, Rockport Publishers, 2010.
4. Chesbrough.H, The era of open innovation, 2003.

Targeted Proficiency and Attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		70	70	70	65	65
Targeted level of Attainment	**Level 3**	60	60	60	60	60
	Level 2	50	50	50	50	50
	Level 1	50	50	50	50	50

Lecture Plan:

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Introduction to OBE, Dissemination of Vision, Mission of the Dept. and PEOs, POs & PSOs of the Programme.		1	Lecture	IFP
2		Identify the elements of design thinking.	K1	1	Lecture	Charts + crayons
3		State the principles of design.	K1	1	Lecture	IFP
4		Discuss the importance and challenges of Design Thinking process.	K2	2	Lecture with discussion	IFP
5		Explain the history of Design Thinking	K1	1	Lecture + Discussion	IFP
6		Explain about new products available in the market	K2	1	Lecture + Discussion	IFP
7		Explain about innovative technologies present in the society	K2	1	Lecture with Discussion and in class Assignment	IFP
		Total		**8**		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO2	Illustrate Design Thinking Process.	K2	1	Lecture	IFP
2		Discuss the importance of Empathy.	K1	1	Discussion	IFP
3		List the problems that are identified around them.	K2	1	Discussion	IFP
4		Identify problem statement	K1	1	Lecture + Discussion	IFP
5		Explain how ideas are to be converted into implementation	K2	1	Lecture + brain storming	IFP
6		Explain the tools involved in Design thinking.	K2	1	Lecture with discussion and in class Assignment	IFP
7		Discuss persona and how to approach end user	K2	2	Lecture	IFP
8		Explain the importance of journey map and product development	K2	2	Lecture	IFP
		Total		**10**		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO3	Define Innovation and its purpose.	K1	1	Lecture	IFP
2		Explain the difference between innovation and creativity	K2	1	Lecture + Discussion	IFP
3		Explain the role of creativity and innovation in an organisation.	K2	1	Lecture	IFP
4		Illustrate the steps from creativity to innovation	K2	2	Lecture+ Discussion	IFP
5		Identify the teams for innovation	K1	1	Lecture + Discussion	IFP
6		Describe the impact and value of creativity	K1	2	Lecture + Discussion	IFP
		Total		**8**		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO4	Show product formation & design	K1	1	Lecture	IFP + Products
2		Demonstrate product design strategies	K2	1	Practical	IFP + Example Products
3		Illustrate the role of product value, and product planning	K2	2	Lecture + Discussion	IFP
4		Explain the importance of modelling.	K2	1	Lecture	IFP + Charts
5		Explain how to set the specifications of a product	K2	2	Lecture + Discussion	IFP
6		Innovative Ideas for their own product	K2	1	Lecture	IFP + charts
7		Case Studies : Detailed study of the needs of the end user and develop a product design	K3	2	Lecture + Discussion + class Assignment	IFP +ICT
		Total		**10**		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	**CO5**	Apply design thinking strategies in Business	K3	1	Lecture	BB+ICT
2		Experiment with innovative ideas to redesign the product	K3	2	Discussion	ICT+ Models
3		Explain the challenges in the business	K2	2	Lecture	BB
4		Analyse the needs of the business	K2	2	Lecture	BB+ICT
5		Develop a prototype for their product.	K3	2	Lecture	BB
6		Motivate the students for start up ideas	K4	2	Lecture with Discussion + Assignment	BB+ICT
		Total		**11**		

Total No. of Classes: 47

Professional Communication Skills - I

Academic Year: 2024-25 **Programme: B.Tech**
Year/ Semester: III **Section: A,B,C& D**
Name of the Course: Professional Communication Skills -III
CourseCode:V23ENT02

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level): After completion of this course, the students will be able to:

S. No.	CO No.	Course Outcome	BTL
1.	CO1	Distinguish the subtle meanings of various words in different contexts, recognize similar words as well as words with contrast meanings and use them appropriately. Express writer's tone and relevant ideas using different types of writing skills and prepare a resume to showcase skills and accomplishments. Organize thoughts in the discussions and express views without reticence. Develop the ability to write different types of essays in a structured way, maintaining cohesion and logic.	K4
2.	CO2	Identify the central theme and arrange the scrambled sentences into a meaningful passage.Draft emails with appropriate subject-lines and relevant content.Compare different pairs of words, recognize the relationship between the head words and the options to siphon correct analogy Choose an appropriate word to make a sentence meaningful.Infer the meaning of the picture by thinking out of the box and speak without inhibitions and face interviews with aplomb.	K3
3.	CO3	Analyze appropriate methods of logical thinking on Ratio and Proportion, Partnership, LCM and HCF, Number System, Areas & Volumes.	K4
4.	CO4	Demonstrate problem solving skills through the concepts of Percentages, Profit and loss, Simple Interest & Compound Interest and Allegation	K3
5.	CO5	Calculate the end results of Cubes, Dice and Data Analysis, Time & Work, Time & Distance, Race & Games.	K4

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	55	55
2	55	55
3	55	55
4	55	55
5	55	55

Text Books:
T1: PIC-VOC
T2: VERBAL ABILITY-2
T3: WORK BOOK ON APTITUDE

Reference Books:

- ❖ Dr.Sujani Tata et al., Pic Voc (2015) – Published by Sri Vasavi Engineering College
- ❖ Lewis Norman, Word Power Made Easy (2008). Goyal Publishers & Distributors Pvt. Ltd.
- ❖ Dr.Shalini Verma, Reetesh Anand, Word Power Made Handy(2017). S Chand Publications.
- ❖ R S Aggarwal,Objective General English (2017). S Chand Publications.
- ❖ Sunita Mishra & C.Muralikrishna, Communication Skills for Engineers (2006). Dorling Kindersley (India) Pvt. Ltd., licensees of Pearson Education in South Asia.
- ❖ Charles W Hanson. Resume: Writing 2020 The Ultimate Guide to Writing a Resume
- ❖ that Lands YOU the Job! (2019).
- ❖ Raymond Murphy. Essential Grammar in Use (1985).Cambridge University Press
- ❖ Seely John. The Oxford Guide to Writing & Speaking (2004). Oxford University Press.
- ❖ Jain,T.S. & Gupta. , 2010, Interviews and Group Discussions, Upkar's Publications.
- ❖ Training & Placement cell, 2020, Workbook -1 on Aptitude, Sri Vasavi Engineering College.
- ❖ M Tyra, 2013, Magical Book on Quicker maths, BSC Publications.
- ❖ K Kundan & M Tyra, 2009, Practice Book on Quicker Maths, BSC Publications.
- ❖ Dr. RS. Agarwal , 2017, Quantitative Aptitude, Sultan Chand Publications
- ❖ Dr. RS. Agarwal, 2017, A modern approach to verbal & on verbal reasoning, Sultan Chand Publications.

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Infer the contextual meaning of words and its contextual usage.	K2	4	Lecture	PPT/A.V
2		Predict the Synonyms-Antonyms of words.	K2	4	Discussion	A.V
3		Develop their resume as per job description	K3	2	Lecture	PPT/A.V
4		Outline paragraphs and essays	K4	2	Lecture	A.V
5		Arrange their ideas logically and present effectively in GDs	K4	2	Lecture	PPT/A.V
6		Connect their views logically and coherently to participate in JAM, and Presentations	K4	3	Lecture	PPT/A.V
7		Relate their creative pursuit to design Advertisements	K4	1	Lecture	PPT/A.V

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Recognize and master "Tenses' and 'Voice' to speak and write effectively.	K1	2	Lecture	PPT/A.V
2		Present their ideas concisely to draft Emails	K1	2	Lecture	PPT/A.V
3		Explain an unfamiliar concept or idea using an analogy.	K2	2	Lecture	PPT/A.V
4		Express their achievements and SWOT confidently to face different types of Interviews	K2	2	Lecture	PPT/A.V
5		Interpret the given picture and write a creative paragraph using writing strategies	K2	2	Lecture	PPT/A.V

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Identify the Next letter or Number in a correct Relation.	K4	2	Lecture	PPT/A.V
2		Justify the relation between words and Numbers.	K4	2	Lecture	PPT/A.V
3		Identify different one from group of terms.	K4	2	Lecture	PPT/A.V
4		Describe their Rank in a class or in a Computation.	K4	2	Lecture	PPT/A.V

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Classify the ages in family members/ Explain the relation between numbers.	K3	2	Lecture	PPT/A.V
2		Calculate the Actual time in Mirror and Water/ Classify the Images.	K3	2	Lecture	PPT/A.V
3		Differentiate the logic behind the conclusions.	K3	2	Lecture	PPT/A.V
4		Explain the logic for a given problem.	K3	2	Lecture	PPT/A.V

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Choose the correct relation between the persons.	K4	2	Lecture	PPT/A.V
2		Show the correct direction.	K4	2	Lecture	PPT/A.V
3		Calculate the Average of data.	K4	2	Lecture	PPT/A.V
4		Relate the correct day for a given date and angle between two hands of a clock.	K4	2	Lecture	PPT/A.V
5		Intercept data.	K4	2	Lecture	PPT/A.V
6		Report the Real Time Scenarios possibility..	K4	2	Lecture	PPT/A.V

Total No. of Classes: 56